Clothing

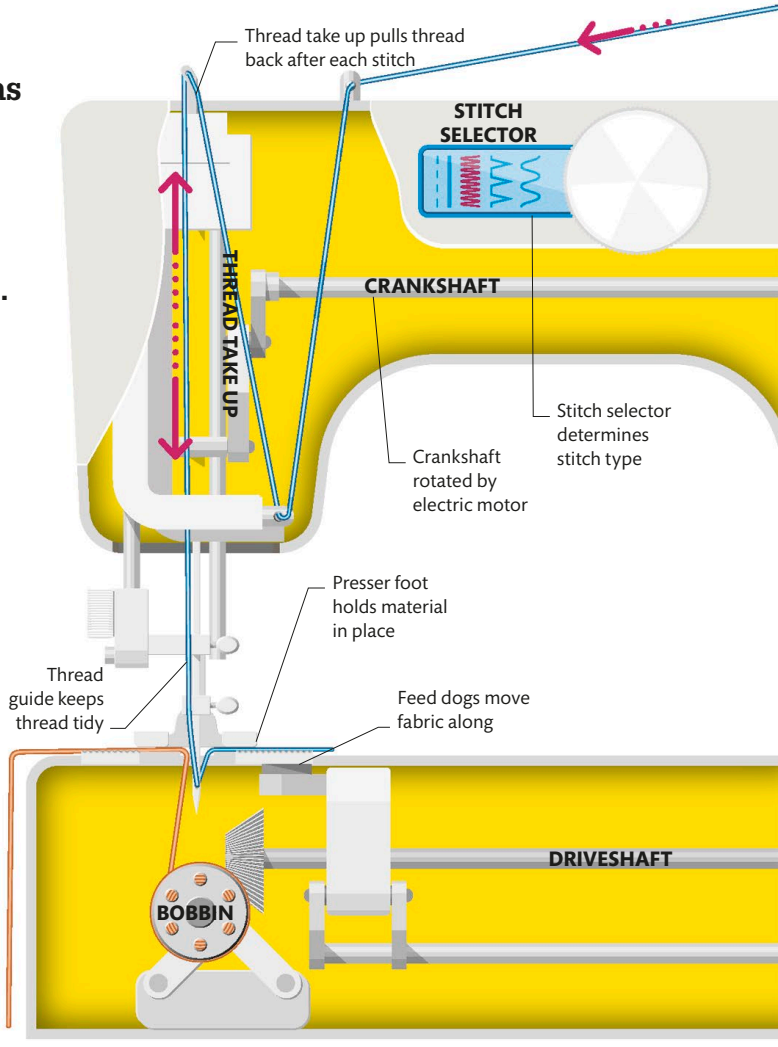
For most of human history, clothing was handmade at home. Even today, when mass-produced clothing dominates most people's wardrobes, some people prefer to create their own garments or make their own alterations and repairs.

Sewing machines

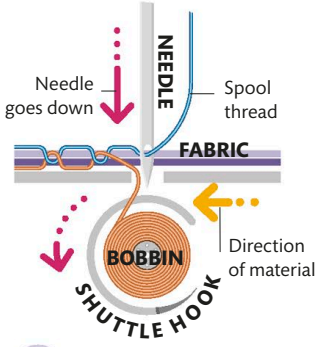
Sewing machines enable fast, accurate stitching to join fabrics together or produce a hem. Thread from a spool is guided through the needle, which moves up and down via a crank turned by the driveshaft. The driveshaft is powered by an electric motor. At the same time, feed dogs move the fabric in a synchronized fashion with the needle to produce a row of equal-sized stitches.

DOMESTIC SEWING MACHINES CAN HAVE A SEWING SPEED OF MORE THAN 1,000 STITCHES PER MINUTE

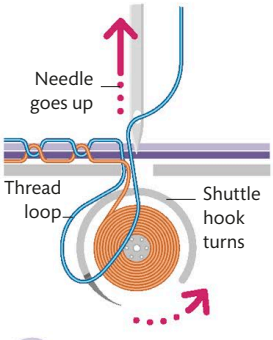




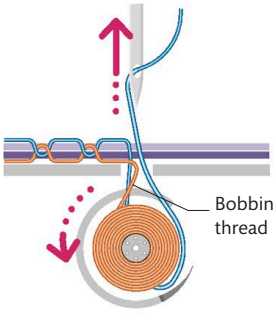
Making a stitch
A household electric sewing machine uses two threads to make a stitch. The machine's controls allow the user to change the size and the type of stitch used on fabric or clothing.



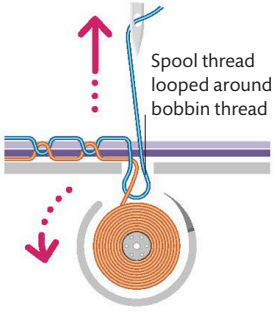
1 Lowering the needle
The needle lowers and penetrates the fabric, carrying the spool thread (blue) down to a spindle of thread, which is called a bobbin (red).



2 Hooking a loop
As the needle travels upward, it leaves a loop of spool thread, which is caught by the shuttle hook as it rotates around the bobbin.



3 Carrying the thread
The shuttle hook carries spool thread around the bobbin case, before the thread slips off the hook and around the bobbin thread.



4 Pulling the stitch
Both threads are pulled up as the needle rises and the fabric advances. The threads are pulled into a stitch, which is tightened by the rising needle.